

UNSW-NB15 Competition Feature Dictionary

ML Hackathon - Network Intrusion Detection | 38 Feature Reference Guide

This dictionary documents the **38 competition features** and **2 ground truth labels** provided in the dataset package. The dataset is derived from UNSW-NB15 with **9 leaky/identifier features removed**. Files train.csv and validation.csv contain 40 columns (38 features + attack_cat + Label). Files challenge.csv and final_test.csv contain 38 columns (features only). All files are headerless.

Competition Features (Columns 0 to 37 + Ground Truth 38, 39)

Idx	Feature Name	Data Type	Description	Example Values
0	proto	Categorical (String)	Transaction protocol (e.g. tcp, udp, arp, ospf, sctp, etc.). 135 unique values.	tcp, udp
1	state	Categorical (String)	Indicates connection state and dependent protocol (e.g. FIN, CON, INT, REQ, RST, ECO, etc.).	FIN, CON, INT
2	dur	Float (Seconds)	Record total flow duration in seconds.	0.001055, 0.036133
3	sbytes	Integer (Bytes)	Source-to-destination transaction bytes.	132, 528, 1464
4	dbytes	Integer (Bytes)	Destination-to-source transaction bytes.	164, 304, 82
5	sloss	Integer (Packets)	Source packets retransmitted or dropped.	0, 1, 5
6	dloss	Integer (Packets)	Destination packets retransmitted or dropped.	0, 2, 8
7	service	Categorical (String)	Application service name: http, ftp, smtp, ssh, dns, ftp-data, irc, pop3, dhcp, ssl, snmp, radius, or '-' if unknown/unused.	dns, http, -
8	Sload	Float (Bits/sec)	Source bits per second transmitted.	500473.9, 87676.1
9	Dload	Float (Bits/sec)	Destination bits per second transmitted.	621800.9, 50480.2
10	Spkts	Integer (Count)	Source-to-destination packet count.	2, 4, 10
11	Dpkts	Integer (Count)	Destination-to-source packet count.	2, 4, 12
12	swin	Integer	Source TCP window advertisement value.	0, 255
13	dwin	Integer	Destination TCP window advertisement value.	0, 255
14	stcpb	Integer	Source TCP base sequence number.	0, 34291823
15	dtcpb	Integer	Destination TCP base sequence number.	0, 19283741
16	smeansz	Integer (Bytes)	Mean packet size transmitted by the source.	66, 132, 512
17	dmeansz	Integer (Bytes)	Mean packet size transmitted by the destination.	82, 76, 1024
18	trans_depth	Integer (Count)	Represents the pipelined depth into connection of HTTP request/response transaction.	0, 1, 2
19	res_bdy_len	Integer (Bytes)	Actual uncompressed content size of data transferred from the server's HTTP service.	0, 1024, 45200
20	Sjit	Float (mSec)	Source jitter in milliseconds.	0.0, 9.891, 14.52
21	Djit	Float (mSec)	Destination jitter in milliseconds.	0.0, 10.68, 8.11
22	Sintpkt	Float (mSec)	Source inter-packet arrival time in milliseconds.	0.017, 7.005
23	Dintpkt	Float (mSec)	Destination inter-packet arrival time in milliseconds.	0.013, 7.564
24	tcprtt	Float (Seconds)	TCP connection setup round-trip time (sum of synack and ackdat).	0.0, 0.00067, 0.12
25	synack	Float (Seconds)	TCP connection setup time between SYN and SYN_ACK packets.	0.0, 0.00035
26	ackdat	Float (Seconds)	TCP connection setup time between SYN_ACK and ACK packets.	0.0, 0.00032
27	is_sm_ips_ports	Binary (0/1)	1 if source and destination IP addresses are equal and port numbers are equal; 0 otherwise.	0, 1
28	ct_flw_http_mthd	Integer (Count)	Number of flows having HTTP methods such as GET and POST.	0, 1, 4
29	is_ftp_login	Binary (0/1)	1 if the FTP session is accessed with user/password; 0 otherwise.	0, 1
30	ct_ftp_cmd	Integer (Count)	Number of flows having a command in FTP session.	0, 1, 2
31	ct_srv_src	Integer (Count)	Number of connections containing the same service and source address in 100 connections.	1, 3, 12
32	ct_srv_dst	Integer (Count)	Number of connections containing the same service and destination address in 100 connections.	1, 7, 24
33	ct_dst_ltm	Integer (Count)	Number of connections of the same destination address in 100 connections according to last time.	1, 2, 8

34	ct_src_ltm	Integer (Count)	Number of connections of the same source address in 100 connections according to last time.	1, 3, 15
35	ct_src_dport_ltm	Integer (Count)	Number of connections of the same source address and destination port in 100 connections.	1, 2, 5
36	ct_dst_sport_ltm	Integer (Count)	Number of connections of the same destination address and source port in 100 connections.	1, 2, 4
37	ct_dst_src_ltm	Integer (Count)	Number of connections of the same source and destination address in 100 connections.	1, 2, 9
38	attack_cat	Categorical (String)	GROUND TRUTH (train/val only): Attack category classification. Categories in initial data: Normal (or empty), Generic, Exploits, Fuzzers, Analysis, Worms.	Generic, Exploits, Normal
39	Label	Binary (0/1)	GROUND TRUTH (train/val only): Primary target class. 0 = Normal, 1 = Attack.	0, 1

Removed Leaky Features (Not Present in Any File)

Feature Name	Original Type	Reason for Removal
srcip	Nominal (IP)	Source IP address. Removed to prevent models from memorizing specific host addresses rather than generalizable traffic patterns.
sport	Integer (Port)	Source port number. Ephemeral client ports leak testbed-specific infrastructure.
dstip	Nominal (IP)	Destination IP address. Target victim IP addresses allow trivial shortcut memorization.
dsport	Integer (Port)	Destination port number. Static service ports lead to artificial shortcut rules.
Stime	Timestamp	Record start timestamp. Attack windows occurred in specific fixed timeframes during capture.
Ltime	Timestamp	Record last timestamp. Strongly correlated with recording session timeframes.
sttl	Integer (TTL)	Source-to-destination Time To Live. In the capture environment, attack traffic originates from VMs with distinctive TTL values, allowing artificial shortcuts without learning traffic dynamics.
dttl	Integer (TTL)	Destination-to-source Time To Live. Reveals testbed hop topography.
ct_state_ttl	Integer	Derived counter based on state and TTL ranges. Encodes the same TTL testbed artifact.